MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2026

FOURTH SEMESTER

DSC A – MG4DSCBTG200 – MOLECULAR BIOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A - One word answer - Answer any 10 questions. Each question carries 2 marks

1. Identify the sugars present in DNA and RNA. [U] / [CO1]
2. List out the types of RNA. [K] /[CO1]
3. Name the scientists who identify split genes. [K] / [CO2]
4. Identify the process by which DNA makes a copy of itself. [U] / [CO3]
5. Recognize start codon and stop codon [K] / [CO3]
6. Name the enzyme that helps in DNA unwinding [K] / [CO3]
7. Recognize the region in which RNA polymerase binds during transcription in prokaryotes [K] / [CO3]
8. Select the RNA that helps in ribosomal assembly during translation. [U] / [CO3]
9. Recall the process fixing UV induced DNA damage in prokaryotes [K] / [CO4]
10. Identify the inducer of Lac operon [K] / [CO4]
11. State the term for jumping genes [K] / [CO4]
12. Expand the terms SINES and LINES [U] / [CO4]

(10X2= 20)

Part B- Short essay- Answer any 5 questions. Each question carries 4 marks

13. Illustrate SOS repair mechanism. [U] / [CO4]
14. State amino acylation for translation. [U] / [CO3]
15. Explain Griffith experiment with diagram. [U] / [CO1]
16. Estimate post transcriptional modification in prokaryotes [U] / [CO3]
17. Summarise gene structure in eukaryotes [U] / [CO2]
18. Discuss DNA polymerases involved in replication. [U] / [CO3]
19. Describe mRNAwith its functions [] / [CO1]

(5X4=20)

Part C- Essay- Answer any 1 question. Each question carries 10 marks

20. Discuss genetic code and its properties [U] /[CO3]

21. Illustrate molecular details of Trp Operon with diagram [U]/[CO4]

(1X10=10)